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Exploring the Feasibility of the Anisotropic Property of ReS₂ in Long-Range Surface Plasmonic Sensing for Superior Performance

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Abstract

This paper presents a novel Kretschmann-configured Long-Range Surface Plasmon Resonance (LRSPR) sensor based on the anisotropic property of Rhenium disulfide (ReS₂) for highly sensitive biomolecular detection. The proposed sensor employs silver (Ag) as the plasmonic metal layer and Lithium niobate (LiNbO₃) as a protective passivation layer to prevent Ag oxidation and enhance stability. The rotation angle is fixed at $\theta = 0^\circ$ to maximise the interaction between incident light and the Re-Re atomic chains of ReS₂. The proposed structure of the sensor is analysed using the Transfer Matrix Method (TMM) in MATLAB, while electromagnetic field distributions and LRSPR modes are investigated through COMSOL Multiphysics simulations. The monolayer ReS₂-based sensor achieves an imaging sensitivity (S_{img}) of 18,361 RIU⁻¹, detection accuracy (DA) of 100/°, angular figure of merit (FOM_{ang}) of 1200 RIU⁻¹, and penetration depth (PD) of 713.11 nm. Further enhancement in S_{img} is observed for bilayer and trilayer of ReS₂-based sensors. Compared with the conventional SPR sensor, the proposed LRSPR sensor demonstrates significantly improved performance parameters, which makes it highly suitable for ultrasensitive detection of biomolecules in advanced biosensing applications.

Keywords: Long-range surface plasmon resonance (LRSPR), Rhenium disulfide (ReS₂), Anisotropic property, Biomolecular sensing, Imaging sensitivity, Lithium niobate (LiNbO₃),

Introduction

The surface plasmon resonance (SPR) technique is widely recognised for its real-time, label-free detection capability and high sensitivity in sensing applications [1],[2],[3]. Its unique operational characteristics and broad applicability distinguish it from other optical sensing technologies [4]. In a conventional SPR (CSPR) sensor, a thin metal layer is deposited over a prism surface. However, CSPR sensors suffer from several limitations, including high propagation losses and a limited ability to detect larger cells or micron-scale biomolecules due to their shallow penetration depth (PD), typically ranging from 100 to 200 nm [5]. The broader resonance curves of

CSPR sensors reduce the detection accuracy (DA) and sensing resolution. To resolve these issues, Long-Range Surface Plasmon Resonance (LRSPR) sensors were introduced. LRSPR sensors provide a significantly larger PD of approximately 500-1000 nm, narrower resonance curves, enhanced DA, and improved angular figure of merit ($FOM_{ang.}$) [6],[7]. Achieving higher sensitivity and resolution is essential for detecting very small refractive index (RI) variations caused by low concentrations of target biomolecules.

The concept of long-range surface plasmon polaritons (LRSPPs) was first reported by Sarid *et al.* in 1981. The LRSPR phenomenon occurs when a thin metal layer is sandwiched between two dielectric layers having nearly identical RI values. Under this condition, surface plasmon polaritons (SPPs) are excited at the interfaces of the metal layer from both sides, producing symmetric and antisymmetric coupling modes that generate long-range and short-range surface plasmon polaritons, respectively. Unlike short-range SPPs, LRSPPs exhibit stronger electric field confinement inside the analyte region and significantly larger PD, extending into the micrometre range [8],[9].

Similar to CSPR sensors, the resonance condition in LRSPR sensors is achieved when wave-vector matching between the incident light and the long-range surface plasmon wave (LRSPW) takes place. Prism-coupled configurations such as Kretschmann and Otto are commonly employed for excitation of plasmon waves. Among these, the Kretschmann configuration is more widely adopted because of its efficient light coupling capability and superior signal-to-noise ratio (SNR) [14],[15]. Consequently, LRSPR sensors demonstrate exceptional sensitivity toward small RI changes in the sensing medium (SM) [11]. Recent advancements in LRSPR technology have primarily focused on improving imaging sensitivity ($S_{img.}$), imaging figure of merit (IFOM), DA, $FOM_{ang.}$ and PD [1] [12],[13].

In 2014, Yang *et al.* reported an LRSPR sensor based on a glass/Cytop/gold multilayer structure for highly sensitive detection of adherent cell fluctuations [16]. In the same year, Zekriti *et al.* experimentally verified that LRSPR sensors provide superior spectral resolution and higher FOM compared to conventional SPR sensors [9]. Later, Singh *et al.* (2021) proposed an antimonene-assisted Kretschmann-configured LRSPR imaging sensor and achieved an $S_{img.}$ of 7028.1 RIU⁻¹, $FOM_{ang.}$ of 335 RIU⁻¹ and DA of 12.5 Deg.⁻¹ [6]. In 2024, Kumar *et al.* developed an LRSPR sensor for early malaria detection with an $S_{img.}$ of 24,775 RIU⁻¹, $FOM_{ang.}$ of 732.91 RIU⁻¹, and DA of 50° [7]. Furthermore, Kumar *et al.* (2025) designed a prism-coupled Kretschmann LRSPR sensor for cortisol detection in saliva and reported an $FOM_{ang.}$ of 276.3 RIU⁻¹, DA of 25°, $S_{img.}$ of 13,931 RIU⁻¹, and IFOM of 348,275 (°RIU)⁻¹ [1]. These studies confirm that Kretschmann-configured LRSPR sensors offer high sensitivity, stability, and practical feasibility [17].

Although several LRSPR sensors based on Graphene, Antimonene, Black phosphorus, MXenes, and other two-dimensional (2D) nanomaterials have been reported in the literature, But, some limitations still remain. Most previously reported studies mainly focused on improving sensitivity, while comparatively less attention has been given to enhancing PD, $S_{img.}$ and DA simultaneously. Moreover, many existing LRSPR sensors employ isotropic nanomaterials, which restrict the exploitation of directional optical properties for sensing enhancement. In addition, the application of anisotropic nanomaterials in LRSPR remains unexplored. Particularly, the use of Rhenium disulfide (ReS₂) in LRSPR sensing architectures has not yet been systematically investigated. Furthermore, comparative investigations involving different metallic layers along with multilayer configurations of ReS₂ are still unavailable in the literature. Therefore, a significant research gap exists in understanding how

the anisotropic optical characteristics of ReS_2 can improve LRSPR sensor performance in terms of sensitivity, DA, FOM_{ang} and PD. ReS_2 is a member of the TMDC family, which possesses unique anisotropic optical and electrical properties because of its distorted low-symmetry crystal structure, which makes it highly attractive for sensing applications [18],[19]. Its polarization-dependent absorption characteristics create new opportunities for the design of advanced LRSPR sensors. ReS_2 exhibits a direct bandgap of approximately 1.43 eV that remains nearly independent of thickness, which enables efficient light absorption and enhanced interaction with analyte molecules [18],[20]. Recent studies have demonstrated the effectiveness of ReS_2 in gas, humidity, and ion sensing applications, especially in field-effect transistor (FET)-based sensors exhibiting directional sensitivity [18], [21]

In this work, a Kretschmann-configured LRSPR sensor incorporating ReS_2 is proposed and investigated using the angular interrogation technique. This work represents the first implementation of ReS_2 in an LRSPR sensing for enhancing S_{img} , DA, FOM_{ang} , and PD compared to conventional and previously reported SPR/LRSPR sensors.

The novelty of the proposed work lies in utilising the anisotropic properties of ReS_2 as a biomolecular recognition element (BRE), which enhances biomolecule adsorption and interaction at the sensor surface. Unlike previous studies, monolayer, bilayer, and trilayer configurations of ReS_2 are systematically analysed in the architecture of the LRSPR sensor. In addition, different plasmonic metals, including silver (Ag), aluminium (Al), and gold (Au), are comparatively investigated to evaluate their influence on the sensor's performance. To further validate the superiority of the proposed design, a detailed comparison with conventional SPR sensors is also carried out for different layers of ReS_2 . A detailed study was also conducted to evaluate the influence of RI changes of the SM on the sensor's performance for different layers of ReS_2 . Finally, the steps for the fabrication of the proposed one are discussed to demonstrate its feasibility for practical and real-world biosensing applications.

2. Proposed Design and Modelling of the sensor

This section shows the structures of both the CSPR and the proposed sensor, which utilize Kretschmann configurations in the angular interrogation technique. The Kretschmann configuration is preferred because of its simplicity, reliability and ease of experimentation if we ignore the bulky setup [4]. Fig. 1(a) depicts the CSPR sensor design, which includes a 2S2G prism, Ag and SM. Fig. 1(b) illustrates the multi-stacked structure of the proposed sensor. The chosen operating wavelength (λ) for both sensors is 633 nm. 2S2G chalcogenide glass is ideal for CSPR/LRSPR sensors in visible and NIR regions due to its wide functional range. In the past decade, 2S2G prisms made from elements like sulfur, selenium, and tellurium have become popular because of their broad transparency. The higher RI of 2S2G prisms helps to match the incident light's wave vector with that of SPs [22]. The RI (n) of 2S2G is expressed in Eq.1 [7], where λ is measured in (μm). Further, a DBL of Cytop has been used as a second layer. The RI of magnesium fluoride (MgF_2), lithium fluoride (LiF) and cytop used for simulations are given in Table 2.

Thereafter, the thin layer of Ag is deposited over cytop as the third layer. In SPR sensors, various metals such as Au, Al, and Ag can be employed to generate SPs. At the wavelength of 633 nm, Collision Wavelength (λ_c) and Plasma Wavelength (λ_p) Values of Au, Al, and Ag have been mentioned in Table 1. The RI of metal (Ag, Al and Au) is computed using the Eq. (2) which is known as Drude–Lorentz model. However, Ag exhibits better accuracy as compared to Au and higher sensitivity than Al. The dielectric function of Ag has a near-zero imaginary part across a broad wavelength range and a greater negative real part than Au, resulting in a narrower resonance peak

that enhances SPR/LRSPR sensor accuracy and sensitivity [23]. Despite these advantages, Ag is prone to oxidation and corrosion. To mitigate this, a 10 nm layer of LiNbO_3 is applied as a fourth layer to provide a passivation effect. LiNbO_3 has a wide band gap (3.8–4.0 eV) [24] and excellent transparency in the visible and near-infrared range, making it optically compatible for use in plasmonic structures with Ag. It forms a physical barrier that prevents oxygen or moisture from reaching the Ag surface, thereby preventing oxidation. Therefore, LiNbO_3 is chemically stable and acts as a protective dielectric layer when coated over silver [24],[25]. The RI of LiNbO_3 was computed according to Eq. (3), and the obtained values are reported in Table 2.

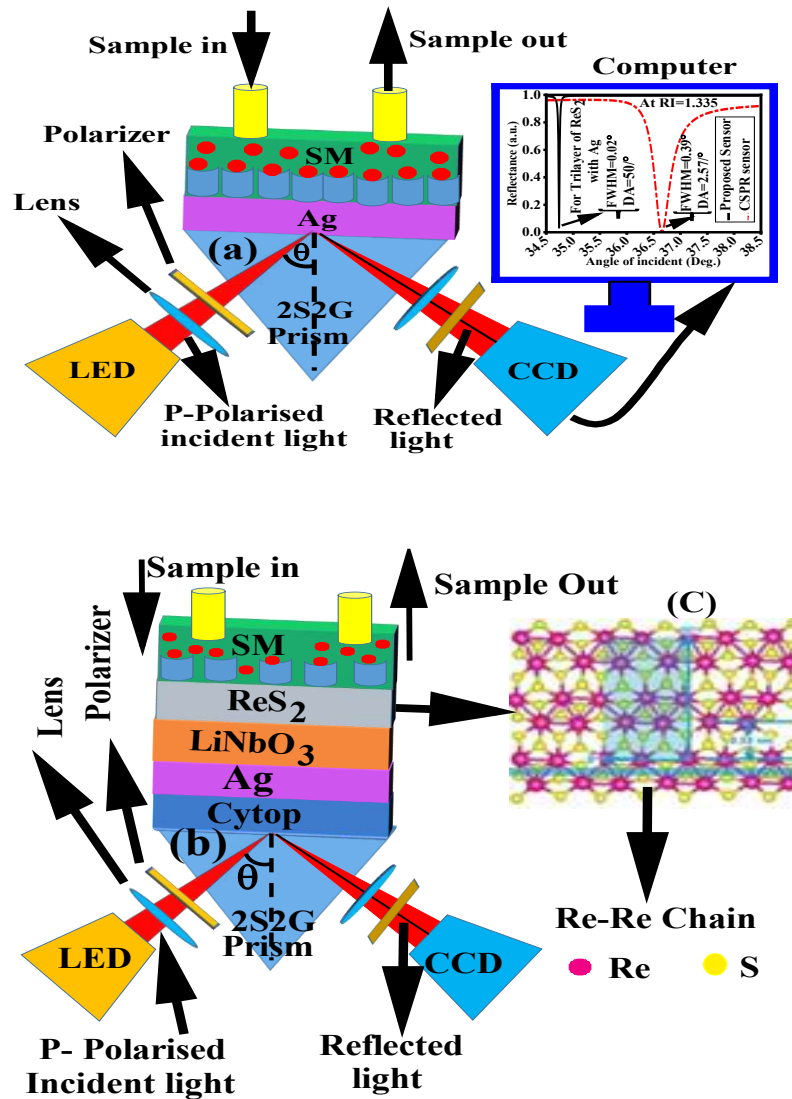


Fig.1 Experimental setup for detecting analyte using (a) C SPR sensor, (b) Proposed sensor, (c) ReS_2 chain [26]

Afterwards, the monolayer, bilayer and tri-layer of ReS_2 is used respectively as fifth layer, which works as the BRE layer to increase the attachment of the desired analyte to the sensor's the surface. The RI of ReS_2 is specifically considered along the direction parallel to the Re-Re atomic chains ($\theta=0^\circ$), where maximum optical absorption and enhanced light-matter interaction have been reported [26],[27]. This directional choice significantly improves SPR coupling and sensor sensitivity, which is essential for designing high-performance

LRSR sensors. The approach follows established anisotropic modelling practices commonly applied in advanced photonic and optoelectronic systems based on ReS₂. The thickness of a ReS₂ monolayer is taken as 0.7 nm [21], and the effective RI for few-layer of ReS₂ is calculated using Eq. (4) [26],[27]. In this equation, θ denotes the rotation angle between the Re-Re chain and the polarisation plane of the incident light. The RIs n_1 and n_2 represent the components parallel and orthogonal to the Re-Re chains, respectively, as shown in Fig. 1(c) [26],[27]. The diagram of the proposed LRSR sensor with ReS₂ layers is shown in Fig. 1(b), and the corresponding RI (n_i) values along the direction parallel to the Re-Re atomic chains are listed in Table 2. The chain orientation can be determined using angle-resolved polarised Raman spectroscopy or transmission spectra analysis [26].

Table 1 At the wavelength of 633 nm Collision Wavelength (λ_c) and Plasma Wavelength (λ_p) Values [30]

S. No.	Metallic layer	Value of (λ_c) in (meter)	Value of (λ_p) in (meter)
1.	Ag	1.7614×10^{-5}	1.4541×10^{-7}
2.	Au	8.9342×10^{-6}	1.6826×10^{-7}
3.	Al	2.4511×10^{-5}	1.0657×10^{-7}

Table 2 The thickness and RIs of the constituent layers at 633 nm.

Layer No.	Materials Used	Optimized Thickness (nm)	RI at 633 nm (0.633 μm .)	Ref. used
1.	2S2G Prism	500	2.358 (A/c to Eq.1)	[7]
2.	LiF	1500	1.391	[8]
	MgF ₂	1500	1.38	[7]
	Cytop	1500	1.34	[7]
3.	Ag	5	A/c to Eq.2	[30]
	Au	5	A/c to Eq.2	[30]
	Al	12	A/c to Eq.2	[30]
4.	LiNbO ₃	10	2.2864 (A/c to Eq.3)	[25]
5.	ReS ₂	1*0.7	4.1937+0.7663i	[26],[27]
		2*0.7	4.1937+0.7965i	[26],[27]
		3*0.7	4.0640+0.7361i	[26],[27]
6.	SM	-	1.33 -1.335	[31]

The SM is treated as the sixth layer in the structure of the proposed one. SM serves as the region where the target samples are introduced for the detection of desired analytes. The RI of the SM in the proposed LRSR sensor is considered to be $1.33 + \Delta n_{SM}$, where $\Delta n_{SM} = 0.005$ signifies the variation in RI caused by the interaction of biomolecules with the ReS₂ layer. When biomolecules present in the sensing sample come into contact with the ReS₂ surface, they get adsorbed, forming a thin biomolecular layer. This adsorption alters the effective RI of the SM to 1.335. This change in RI due to analyte binding provides a shift in the resonance angle (θ_r), which directly indicates the sensitivity of the sensor [28],[29].

$$n(\lambda)_{2S2G} = 2.24047 + \frac{2.693 \times 10^{-2}}{\lambda^2} + \frac{8.08 \times 10^{-3}}{\lambda^4} \quad (1)$$

$$n_{Metal}(\lambda) = \left(1 - \frac{\lambda^2 \times \lambda_C}{\lambda_P^2 (\lambda_C + i\lambda)}\right)^{\frac{1}{2}} \quad (2)$$

$$n_{LiNbO_3} = \sqrt{1 + \frac{2.6734\lambda^2}{\lambda^2 - 0.01764} + \frac{1.2290\lambda^2}{\lambda^2 - 0.05914} + \frac{12.614\lambda^2}{\lambda^2 - 474.60}} \quad (3)$$

$$\frac{1}{n_{ReS_2}^2} = \frac{\cos^2 \theta}{n_1^2} + \frac{\sin^2 \theta}{n_2^2} \quad (4)$$

2.2 Mathematical Modelling and Performance Parameters

In this sub-section, we utilized the Transfer Matrix Method (TMM) and Fresnel equations to find the reflection coefficient for P-polarized TM incident light. TMM is known for its high accuracy as it eliminates the need for approximations. Angular interrogation technique is used to analyse reflection intensity in terms of incident angle for the proposed 6-layer modelled sensor. Each layer in the z-axis arrangement is denoted by its thickness (d_k), dielectric constant (ϵ_k), and RI (n_k). After calculating the reflection coefficient (r_p), the reflectivity (R_p) can be obtained by Eq. (5). For a detailed explanation of this modelling method, cited references of [8],[30],[22],[32],[33] may be referred.

$$R_p = |r_p|^2 \quad (5)$$

$$\text{Where, } r_p = \frac{(M_{11} + M_{12} q_6) q_1 + (M_{21} + M_{22} q_6)}{(M_{11} + M_{12} q_6) q_1 - (M_{21} + M_{22} q_6)} \quad (6)$$

The performance parameters of the proposed one are obtained using the reflectance curves (RC) of LRSPR which were generated through simulations in MATLAB [8],[30],[34]. The main performance parameters for the proposed one are outlined below.

$$\text{Angular Sensitivity, } S_{ang.} = \frac{\Delta \theta_{res}}{\Delta n_{SM}} [^\circ/RIU] \quad (7)$$

$$\text{Imaging Sensitivity, } S_{img.} = \frac{\Delta R_{ref.}}{\Delta n_{SM}} [RIU^{-1}] \quad (8)$$

$$\text{Detection accuracy, } DA = \frac{1}{FWHM} [1^\circ] \quad (9)$$

$$\text{Angular Figure of Merit, } FOM_{ang} = \frac{S_{ang.}}{FWHM} = S_{ang.} \times DA [1/RIU] \quad (10)$$

$$\text{Imaging Figure of Merit, } IFOM = \frac{S_{img.}}{FWHM} = S_{img.} \times DA [1/(^\circ RIU)] \quad (11)$$

Here, $\Delta \theta_{res.}$ indicates shifts in resonance angle, (Δn_{SM}) shows the variation in RI of SM and $\Delta R_{ref.}$ specifies the change in reflectance. $S_{ang.}$ denotes the angular sensitivity.

3. Results and discussion:

3.1 Optimisation of plasmonic metal

In this section, the optimisation of plasmonic metals is carried out by comparing the reflectance curves (RC) in terms of incident angle for different metal (Ag, Al and Au) layers integrated with various layers of ReS₂ as detailed in Table 2. Fig. 2 displays the RC of LRSPR, examining the sensor's response before biomolecule adsorption (At RI = 1.33, pure water) and after adsorption (At RI = 1.335, impure water). This analysis highlights the effect of biomolecular interaction on the sensing surface and assists in identifying the most suitable plasmonic metal to combine with monolayer, bilayer, and tri-layer of ReS₂ along with LiNbO₃, in the structures of the proposed sensor.

3.1.1 Comparison of Reflectance for Various Layers of ReS₂ with Al

In this subsection, a comparative analysis of reflectance has been conducted by examining the RC in terms of the incident angle for aluminium (Al) combined with monolayer, bilayer, and tri-layer of ReS₂.

Fig. 2(a) displays the RC for the monolayer of ReS₂ at RI of 1.33 and 1.35, shown by the dashed red and solid black lines, respectively. The proposed sensor demonstrates a minimum reflectance ($R_{\min}$) of 3.694×10^{-4} a.u., a full width FWHM of 0.02° , and DA of 50° for the SM with RI = 1.335. Fig. 2(b) illustrates the RC for the bilayer of ReS₂ under the same RIs, also denoted by dashed red and solid black lines. Here, the sensor achieves an $R_{\min}$ of 7.531×10^{-2} a.u., FWHM of 0.02° , and DA of 50° at RI = 1.335.

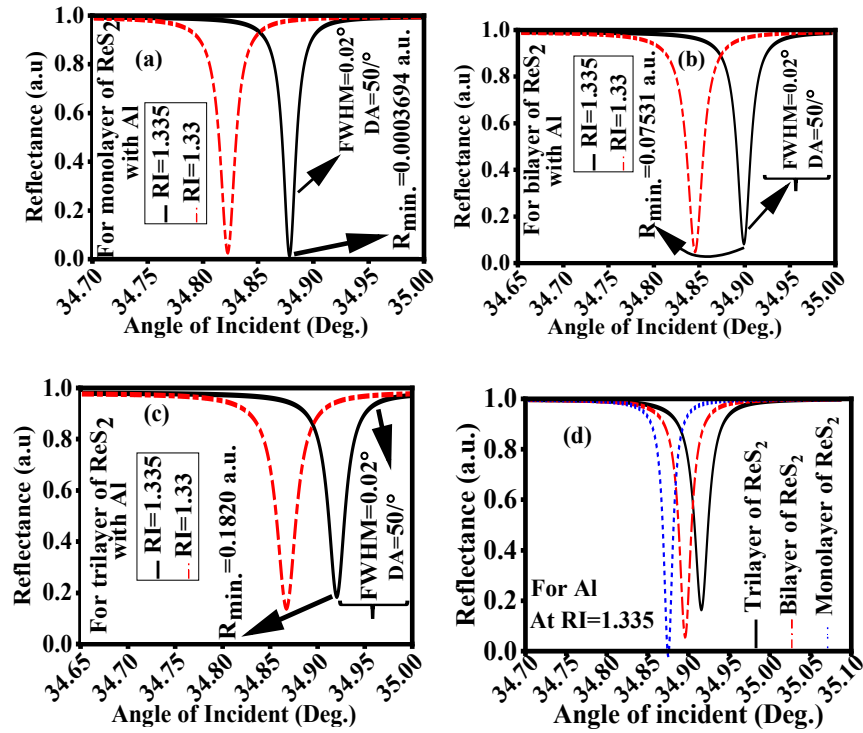


Fig. 2 Optimization of Plasmonic metal (Al) and their thickness in terms of $R_{\min}$ and DA (a) Comparison of RC at RI=1.33 and 1.335 for monolayer of ReS₂ (b) Comparison of RC at RI=1.33 and 1.335 for bilayer of ReS₂ (c) Comparison of RC at RI=1.33 and 1.335 for Trilayer of ReS₂ (d) Comparison of RC at RI=1.335 for mono, bi and trilayer of ReS₂.

Fig. 2(c) presents the results for the tri-layer of ReS₂ at RI = 1.33 and RI = 1.335, again marked by dashed red and solid black lines. The sensor exhibits an $R_{\min}$ of 1.82×10^{-1} a.u., with FWHM and DA remaining consistent at 0.02° and 50° , respectively. Fig. 2(d) provides a direct comparison of the RC at RI = 1.35 for the monolayer, bilayer, and trilayer of ReS₂ configurations, shown by dotted blue, dashed red, and solid black lines, respectively. It is evident from the comparison that the monolayer of ReS₂ combined with Al offers the lowest $R_{\min}$ of 3.694×10^{-4} a.u., which indicates the enhanced absorption of incident light within the structure of the sensor. This significant reduction in reflectance highlights a more efficient light-matter interaction, which is crucial for improved sensor performance [35], [7].

3.1.2 Comparison of Reflectance for Various Layers of ReS₂ with Au

In this subsection, a comparative analysis of reflectance is performed by evaluating the RC in terms of the incident angle for gold (Au) combined with monolayer, bilayer, and tri-layer of ReS₂. Fig. 3(a) shows the RC for the monolayer of ReS₂ at RI of 1.33 and 1.35, represented by dashed red and solid black lines, respectively. For the SM with RI = 1.335, the proposed one achieves an $R_{\min.}$ of 2.002×10^{-1} a.u., a FWHM of 0.01° , and a DA of 100° . Fig. 3(b) presents the reflectance response for the bilayer of ReS₂ under the same RIs, indicated by dashed blue and solid pink lines. In this case, the sensor records an $R_{\min.}$ of 5.33×10^{-2} a.u., FWHM of 0.02° , and DA of 50° for RI = 1.335.

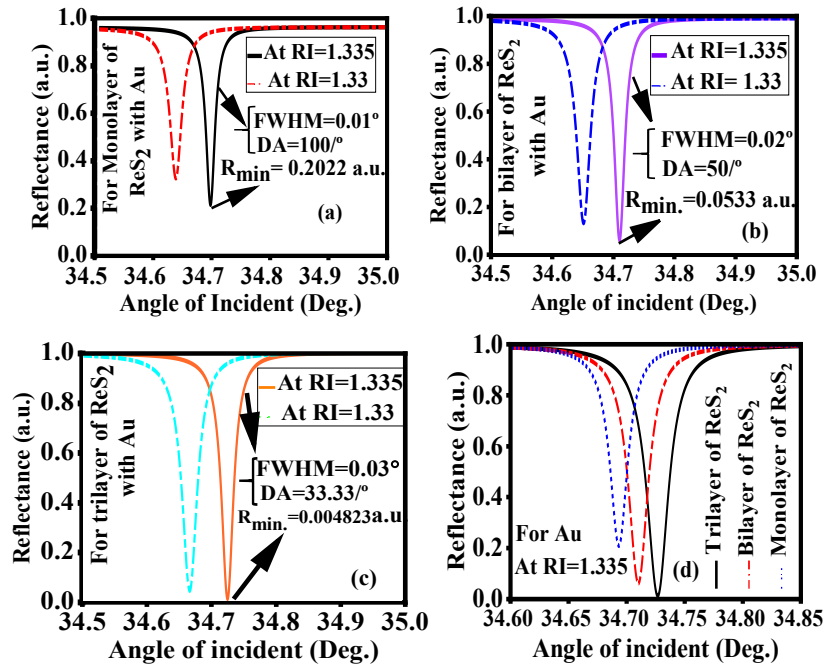


Fig. 3 Optimization of Plasmonic metal (Au) and their thickness in terms of $R_{\min}$ and DA (a) Comparison of RC at RI=1.33 and 1.335 for monolayer of ReS₂ (b) Comparison of RC at RI=1.33 and 1.335 for bilayer of ReS₂ (c) Comparison of RC at RI=1.33 and 1.335 for Trilayer of ReS₂ (d) Comparison of RC at RI=1.335 for mono, bi and trilayer of ReS₂.

Fig. 3(c) illustrates the results for the trilayer of ReS₂ at RI = 1.33 and RI = 1.335, marked by dashed cyan and solid orange lines. The sensor shows an $R_{\min.}$ of 4.823×10^{-3} a.u., with corresponding FWHM of 0.03° and DA of 33.33° . Fig. 3 (d) offers a direct comparison of the RC at RI = 1.35 for monolayer, bilayer, and tri-layer ReS₂ structures, shown by dotted blue, dashed red, and solid black lines, respectively. Among the configurations, the trilayer of ReS₂ combined with Au delivers the lowest $R_{\min.}$ of 4.823×10^{-3} a.u., suggesting superior absorption of incident light within the sensor architecture. However, it is also notable that the monolayer configuration yields the highest DA of 100° , highlighting a trade-off between sensitivity and precision in sensor design.

3.1.3 Comparison of Reflectance for various Layers of ReS₂ with Ag

In this subsection, a comparative analysis of reflectance is conducted by examining the RC in terms of the incident angle for silver (Ag) combined with monolayer, bilayer, and trilayer of ReS₂.

Fig. 4(a) depicts the RC for the monolayer of ReS₂ at RI of 1.33 and 1.35, represented by dashed red and solid black lines, respectively. For the SM with RI = 1.335, the proposed one achieves an $R_{\min}$ of 3.761×10^{-1} a.u., FWHM of 0.01° , and a DA of 100° . Fig. 4(b) presents the reflectance response for the bilayer of ReS₂ under the same RIs, indicated by dashed blue and solid pink lines. In this case, the sensor exhibits an $R_{\min}$ of 1.393×10^{-1} a.u., with FWHM and DA values of 0.01° and 100° , respectively.

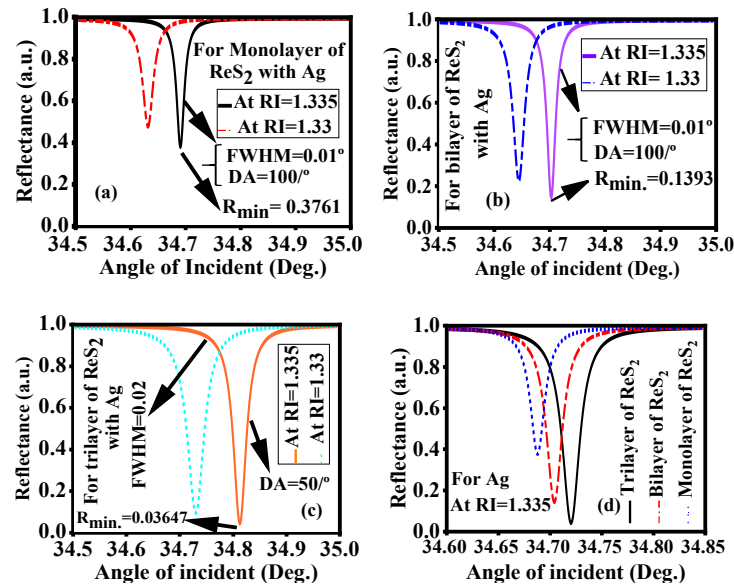


Fig. 4 Optimization of Plasmonic metal (Ag) and their thickness in terms of $R_{\min}$ and DA (a) Comparison of RC at RI=1.33 and 1.335 for monolayer of ReS₂ (b) Comparison of RC at RI=1.33 and 1.335 for bilayer of ReS₂ (c) Comparison of RC at RI=1.33 and 1.335 for Trilayer of ReS₂ (d) Comparison of RC at RI=1.335 for mono, bi and trilayer of ReS₂.

Fig. 4(c) demonstrates the RC for the trilayer of ReS₂ at RI = 1.33 and RI = 1.335, marked by dashed cyan and solid orange lines. The corresponding sensor performance shows an $R_{\min}$ of 3.647×10^{-2} a.u., with a FWHM of 0.02° and DA of 50° . Fig. 4(d) offers a direct comparison of the RC at RI = 1.35 for the monolayer, bilayer, and trilayer configurations of ReS₂, shown by dotted blue, dashed red, and solid black lines, respectively. Among these configurations, the trilayer of ReS₂ combined with Ag delivers the lowest $R_{\min}$ of 3.647×10^{-2} a.u., indicating stronger absorption of incident light and enhanced light-matter interaction, both of which are critical for high-performance sensing.

It is evident from the numerical analysis that as the ReS₂ layer thickness increases from monolayer to trilayer, the sensor performance improves in terms of achieving lower $R_{\min}$ values. However, it is also noteworthy that the monolayer and bilayer configurations yield the highest detection accuracy of 100° , suggesting a trade-off between sensitivity and angular precision.

Based on this comprehensive evaluation, it is clear that the proposed structure of the sensor with Ag, in combination with monolayer, bilayer, and trilayer of ReS₂, exhibits excellent performance characteristics, with each configuration offering specific advantages in terms of $R_{\min}$ and detection accuracy.

Based on the simulation results obtained using in MATLAB are presented in the comparative Tables 3, Table 4 and Table 5, Ag clearly emerges as the top plasmonic metal for the proposed one across all three monolayer, bilayer, and trilayer of ReS₂.

In the case of the monolayer of ReS₂, Ag exhibits superior performance metrics, including an R_{min.} of 0.3761 a.u., a sharp resonance with a FWHM of 0.01°, and the highest DA of 100/°, which it shares with Au. Importantly, Ag achieves the highest S_{img.} of 18,361 RIU⁻¹ and an IFOM of 1.8361×10⁶, outperforming both Au and Al. Although Al offers the lowest R_{min.}, its broader FWHM and lower FOM_{ang.} reduce its overall effectiveness.

Moving to the design of the sensor with a bilayer of ReS₂, Ag maintains its advantage by preserving the lowest FWHM (0.01°) and highest DA (100/°), while also achieving the highest FOM_{ang.} of 1,200 RIU⁻¹. Its S_{img.} of 20,320 RIU⁻¹ and IFOM of 2.032 × 10⁶ further reinforce its superior performance, establishing a clear lead over Au and Al, which show lower S_{ang.} and broader resonance curves.

In the sensor designed with the Tri layer of ReS₂, Ag continues to deliver balanced and effective performance. While the FWHM slightly increases to 0.02°, it still matches to that of Al and remains sharper than FWHM of 0.03° of Au. The S_{ang.} of 12°/RIU, FOM_{ang.} of 750 RIU⁻¹, and S_{img.} of 22,237 RIU⁻¹ are all at the maximum end of the performance spectrum.

Ag offers the highest IFOM value with 1.11185 × 10⁶, indicating strong sensing efficiency. Despite Au achieves the lowest R_{min.} in the trilayer case, its poor FWHM and DA undermine its practical utility. Overall, across all ReS₂ thickness configurations, Ag consistently provides the best combination of high sensitivity, sharp resonance, strong light–matter interaction, and precision. These comprehensive findings confirm that Ag is the most effective plasmonic material for integration with ReS₂ in the proposed design of the LRSPP sensor, making it the ideal choice for high-performance biosensing and environmental detection applications.

Table 3: Comparison of Parameters of the Proposed Sensor for Different Metals with a monolayer of ReS₂

Performance parameter of the proposed one with a monolayer of ReS ₂									
Metal	Thickness(nm)	FWHM(°)	DA(1/°)	S _{ang.} (°/RIU)	FOM _{ang.} (RIU ⁻¹)	R _{min.} (a.u.)	θ _r (°)	S _{img.} (RIU ⁻¹)	IFOM [(°RIU)] ⁻¹
Au	5	0.01	100	12	1200	0.2022	34.69	14871	1487100
Al	12	0.02	50	12	600	0.0003694	34.88	21159	1457950
Ag	5	0.01	100	12	1200	0.3761	34.69	18361	1836100

Table 4: Comparison of Parameters of the Proposed Sensor for Different Metals with a bilayer of ReS₂

Performance parameters of the proposed one with a bilayer of ReS ₂									
Metal	Thickness(nm)	FWHM(°)	DA(1/°)	S _{ang.} (°/RIU)	FOM _{ang.} (RIU ⁻¹)	R _{min.} (a.u.)	θ _r (°)	S _{img.} (RIU ⁻¹)	IFOM [(°RIU)] ⁻¹
Au	5	0.02	50	12	600	0.0533	34.71	23431	1171550
Al	12	0.02	50	10	500	0.07531	34.90	22612	1130600
Ag	5	0.01	100	12	1200	0.1393	34.70	20320	2032000

Table 5: Comparison of Parameters of the Proposed one for Different Metals with a trilayer of ReS₂

Performance parameters of the proposed one with Trilayer of ReS ₂									
Metal	Thickness(nm)	FWHM(°)	DA(1/°)	S _{ang.} (°/RIU)	FOM _{ang.} (RIU ⁻¹)	R _{min.} (a.u.)	θ _r (°)	S _{img.} (RIU ⁻¹)	IFOM [(°RIU)] ⁻¹
Au	5	0.03	33.34	12	400	0.004823	34.73	23986	799533.34
Al	12	0.02	50	10	500	0.1820	34.92	18523	926150
Ag	5	0.02	50	12	750	0.03647	34.72	22237	1111850

3.1.4 Variation of Sensor Performance Parameters with Change in RI of the SM:

The obtained results in the previous subsection clearly demonstrate that Ag is the most suitable plasmonic metal for the proposed ReS₂-based LRSPR sensor due to its strong plasmonic response, lower optical loss, and sharper resonance characteristics. The incorporation of monolayer ReS₂ significantly boosts the interaction between the evanescent field and the SM, which results in improved sensing performance.

For the monolayer ReS₂-based LRSPR sensor, the $S_{\text{img.}}$ increases with the RI change of the SM. As the RI changes from 1.330 to 1.335, the $S_{\text{img.}}$ improves from 12,442 RIU⁻¹ to 18,361 RIU⁻¹ as presented in Table 6 and Fig.5(a). This enhancement occurs because higher RI values strengthen the coupling between the SPW and the analyte layer, leading to a larger shift in θ_r and stronger plasmon–matter interaction.

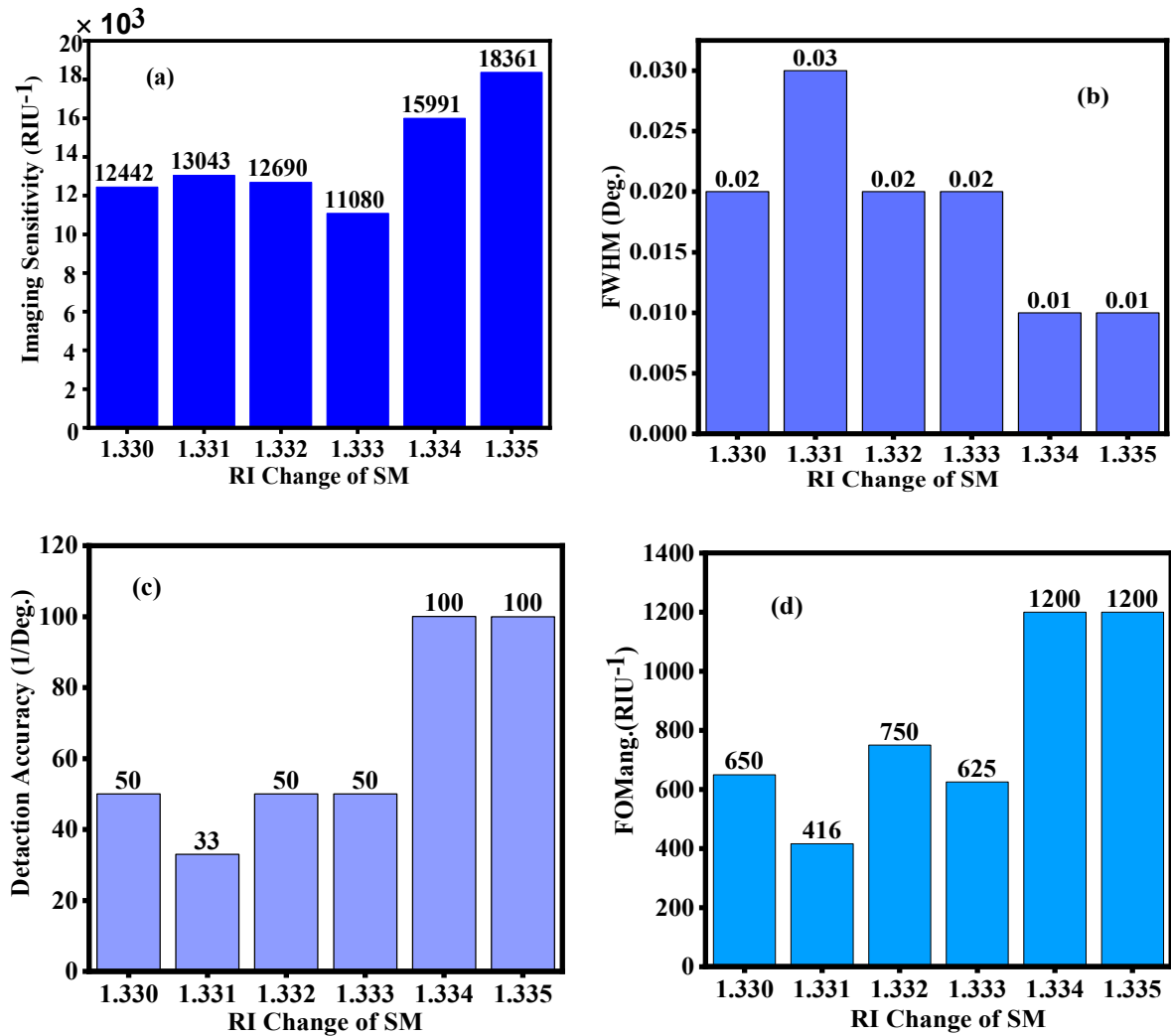


Fig. 5 Variation of Performance parameters of the sensor based on a monolayer of ReS₂ with respect to RI change of SM (a)Imaging Sensitivity vs RI change of SM (b) FWHM vs RI change of SM (c) Detection Accuracy vs RI change of SM (d) FOM_{ang.} vs RI change of SM

The FWHM remains very narrow, varying between 0.01° and 0.03°. A narrow FWHM indicates a sharp resonance dip, which is a desirable characteristic for high-resolution biosensing applications. At RI values of 1.334 and

1.335, the FWHM becomes minimum (0.01°), which shows excellent resonance sharpness and improved detection capability as displayed in Table 6 and Fig. 5(b).

Table 6: Comparison of Parameters of the Proposed Sensor with respect to RI change of SM with various layers of ReS₂

For Monolayer of ReS ₂					For a Bilayer of ReS ₂				For a Trilayer of ReS ₂			
RI Change of SM	FWHM ($^\circ$)	DA ($1/^\circ$)	S _{img.} (RIU ⁻¹)	FOM _{ang.} (RIU ⁻¹)	FWHM ($^\circ$)	DA ($1/^\circ$)	S _{img.} (RIU ⁻¹)	FOM _{ang.} (RIU ⁻¹)	FWHM ($^\circ$)	DA ($1/^\circ$)	S _{img.} (RIU ⁻¹)	FOM _{ang.} (RIU ⁻¹)
1.33	0.02	50	12442	650	0.02	50	14972	600	0.02	58	17950	600
1.331	0.03	33	13043	416	0.02	50	17706	625	0.02	50	17451	600
1.332	0.02	50	12690	750	0.02	50	19942	667	0.02	50	16600	625
1.333	0.02	50	11080	625	0.01	100	20328	600	0.01	50	21710	667
1.334	0.01	100	15991	1200	0.02	50	17994	500	0.02	100	24214	667
1.335	0.01	100	18361	1200	0.01	50	20320	1200	0.02	50	22237	750

In a similar manner, the performance of bilayer and trilayer ReS₂-based LRSPR sensors can also be explained. As the number of ReS₂ layers increases, the interaction between the plasmonic field and the analyte becomes stronger, which further improves the imaging sensitivity from 20,320 RIU⁻¹ (bilayer) to 22,237 RIU⁻¹ (trilayer) as shown in Table 5. However, the addition of extra ReS₂ layers slightly broadens the resonance curve, causing a small reduction in DA and PD. Despite this slight trade-off, the bilayer and trilayer structures still provide substantially better sensing performance compared to the CSPR sensor due to stronger field confinement and enhanced plasmon propagation characteristics in the LRSPR configuration.

3.2 Electric Field Distribution across Layer Interfaces in CSPR and Proposed LRSPR Sensors

In this section, the electric field (EF) distribution at the layer interfaces of both the proposed one and the CSPR sensor was analysed using COMSOL Multiphysics software. The EF distribution analysis, as illustrated in Figs. 6 to 9, clearly demonstrates that the proposed sensor integrated with monolayer, bilayer, and tri-layer of ReS₂ significantly outperforms the CSPR sensor in every critical performance metric. In the CSPR sensor, shown in Fig. 6 (f), the peak electric field (E_y) reaches only about 3.1×10^5 V/m with a PD of 225.14 nm, as shown in Fig. 6 (a). This limited field strength and shallow PD result in weaker interaction with the SM, thereby reducing the sensor's sensitivity [1]. In contrast, the proposed sensor integrated with ReS₂ layers exhibits much stronger electric field confinement and deeper PD. With the monolayer ReS₂, as seen in Fig. 7(f), the peak electric field reaches 6.81×10^5 V/m, more than double that of the CSPR sensor, while the PD extends significantly to 713.11 nm as shown in Fig. 7 (a). The 2D and 3D field distribution plots in Figs. 7(c) and 7(d) show clearly localized and intensified Surface Plasmon Polariton (SPP) modes, indicating a stronger light-matter interaction [35] [1] [7]. Similarly, with the bilayer of ReS₂, shown in Figs.8, The proposed sensor maintains a high field strength of 6.07×10^5 V/m with a PD of 676.44 nm, offering consistent performance improvement with well-confined SPP excitation, as seen in Figs. 8(c) and 8(d) compared to localized and intensified SPP modes in the CSPR sensor as shown in Figs. 6 (c) and 6 (d).

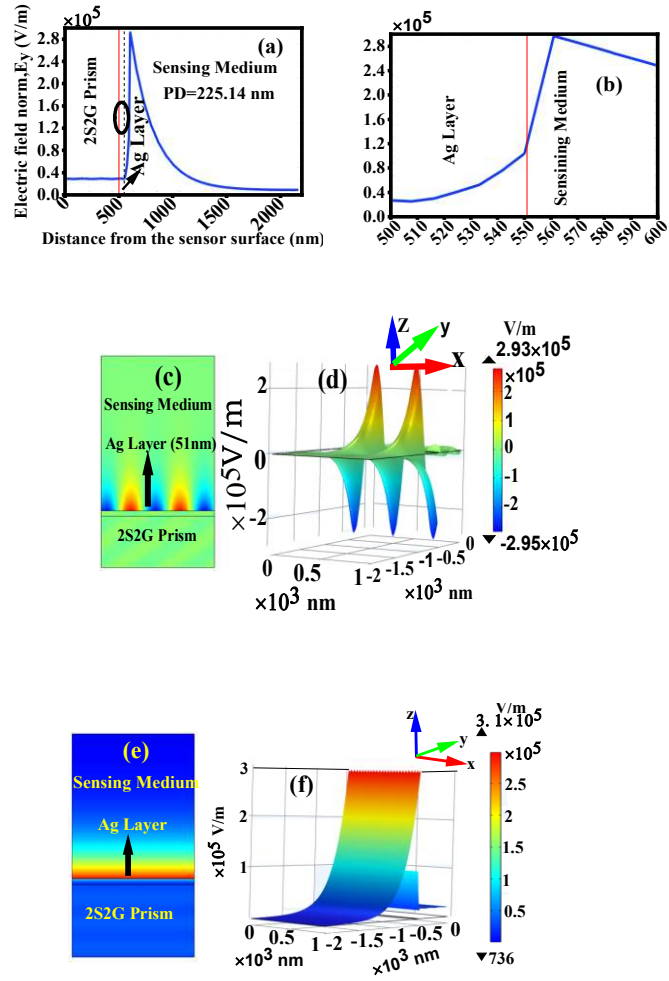
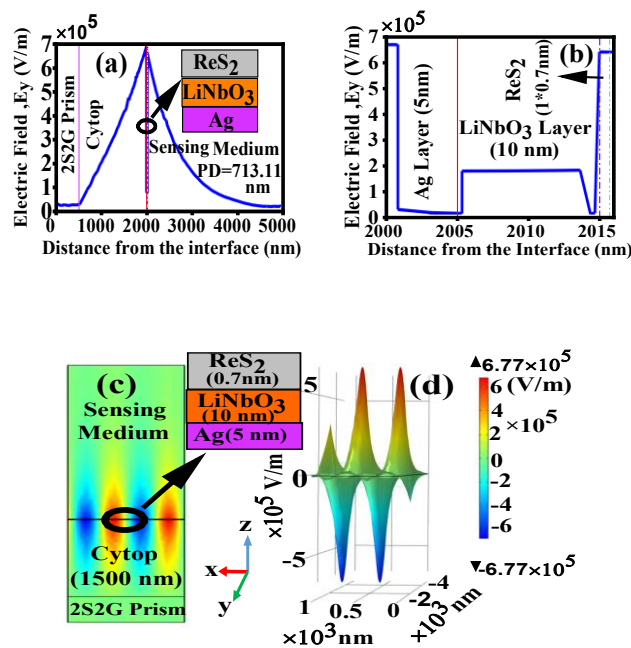


Fig.6 Field distribution in CSPR sensor during detection of analyte at θ_i of 36.66° (a)1D plot of EF variation vs Normal distance from the interface (b) An amplified view of an elliptical area in Fig. 6(a). (c) 2D plot of EF distribution (D) 3D plot of EF distribution (e) 2D Plot for SPP (f) 3D plot of SPP.



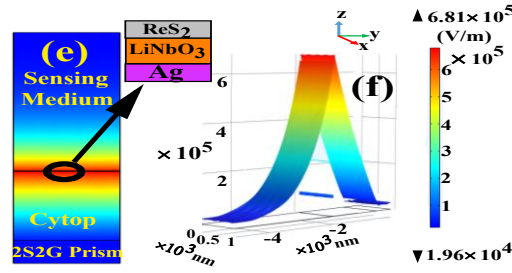
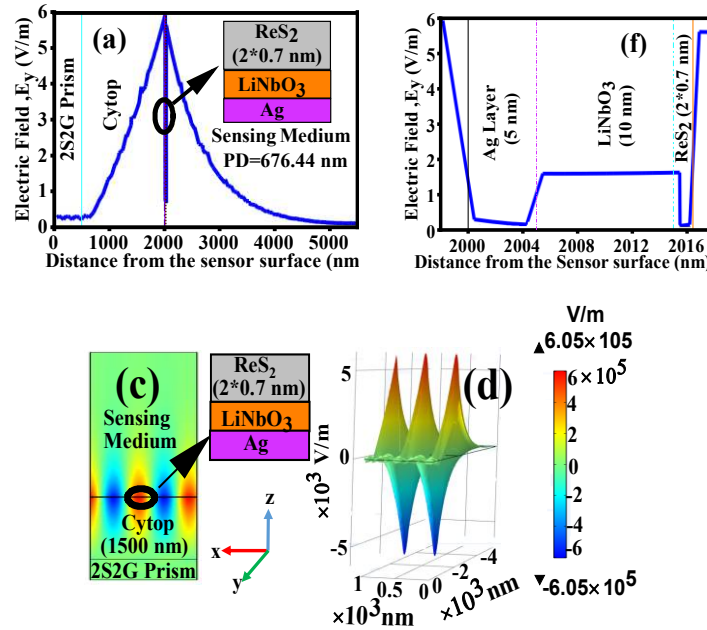


Fig.7 Field distribution in proposed sensor with monolayer of ReS₂ during detection of analyte at θ_r of 34.69 (a) 1D electric field variation vs Normal distance from the sensor surface (b) An amplified view of an elliptical area in Fig. 7(a). (c) 2D plot of EF distribution (D) 3D plot of EF distribution (e) 2D Plot for SPP (f) 3D plot of SPP.

The tri-layer of ReS₂ further reinforces the sensor's superiority. As depicted in Fig. 9, the electric field peaks at 6.41×10^5 V/m with a PD of 644.22 nm, and the field confinement remains sharp and uniform, as displayed in Figs. 9(c) and (d). These values are significantly greater than those observed in the CSPR sensor, confirming that the proposed structure enables enhanced field intensity and deeper analyte interaction across all configurations. Moreover, the inclusion of the LiNbO₃ layer and the anisotropic ReS₂ layer not only improves optical compatibility but also contributes to stronger field enhancement and sensitivity. The field plots in Figs. 7 to 9 [7(e–f), 8(e–f), and 9(e–f)] highlight how well the SPP modes are confined at the interface, further justifying the superior performance of the proposed sensor. Therefore, based on these observations, it is obvious that the proposed one with monolayer, bilayer, or trilayer of ReS₂ demonstrates significantly stronger EF generation, deeper penetration into the SM, and enhanced SPP excitation compared to the CSPR sensor. These improvements in results makes the sensor more sensitive and efficient which makes it a clearly better choice for high-performance plasmonic sensing applications under any condition.



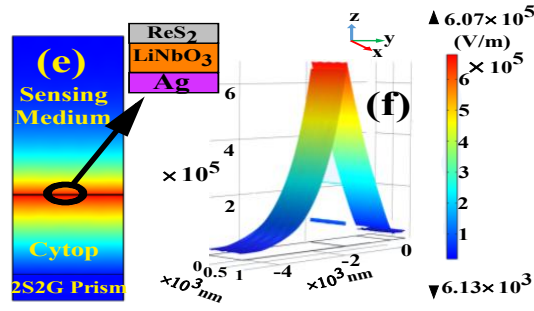


Fig.8 Field distribution in proposed sensor with bilayer of ReS_2 during detection of analyte at θ_r of 34.70 (a) 1D EF variation vs Normal distance from the sensor surface (b) An amplified view of an elliptical area in Fig. 8(a). (c) 2D plot of EF distribution (D) 3D plot of EF distribution (e) 2D Plot for SPP (f) 3D plot of SPP.

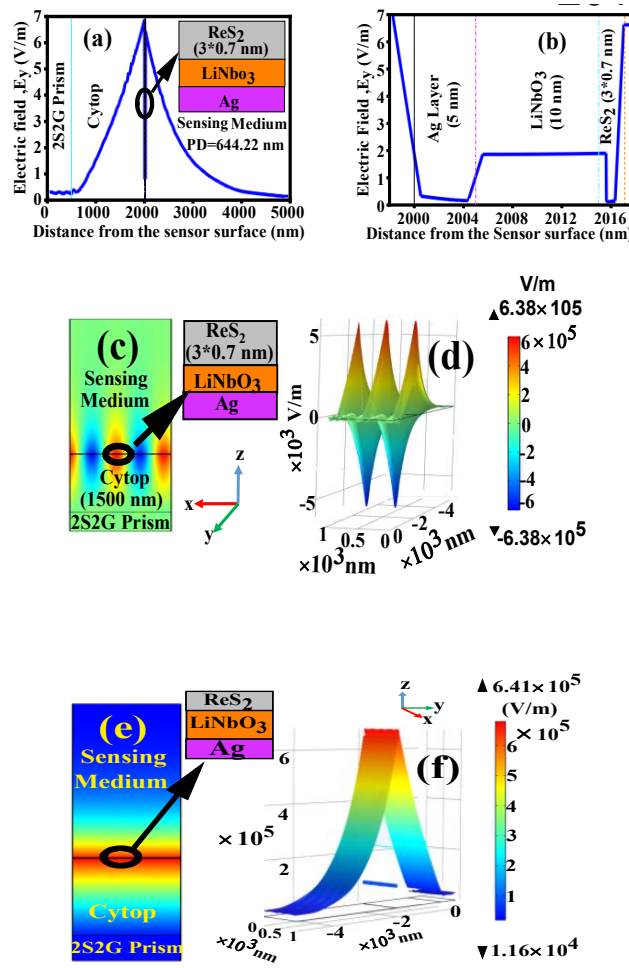


Fig.9 Field distribution in proposed sensor with trilayer of ReS_2 during detection of analyte at θ_r of 34.72 (a) 1D EF variation vs Normal distance from the sensor surface (b) An amplified view of an elliptical area in Fig. 9(a). (c) 2D plot of EF distribution (D) 3D plot of EF distribution (e) 2D Plot for SPP (f) 3D plot of SPP.

5. Comparison of the performance parameters of CSPR and the proposed sensor

Based on Figs.10 and the data provided in Table 7, it is evident that the proposed sensor, incorporating monolayer, bilayer, and tri-layer of ReS₂ with Ag, significantly outperforms the CSPR sensor in all performance metrics. This comparative analysis justifies that the proposed one is far superior for plasmonic sensing applications.

In Fig. 10 (a), the CSPR sensor shows a broader reflectance dip with a FWHM of 0.39° and a DA of just 2.57°. In comparison, the proposed one, as demonstrated in Fig. 10(b), 10(c), and 10 (d) for monolayer, bilayer, and trilayer of ReS₂ respectively, exhibits extremely sharp resonance curves. The monolayer and bilayer configurations both achieve narrow FWHM of 0.01° and DA of 100°, while the trilayer still performs with an FWHM of 0.02° and DA of 50°. This indicates a drastic improvement in angular resolution and precision in analyte detection. From Table 6, the proposed sensor also demonstrates significant gains in $FOM_{ang.}$, where the monolayer and bilayer versions reach 1200 RIU⁻¹, compared to only 92.31 RIU⁻¹ for the CSPR sensor. Moreover, the $R_{min.}$ values, is numerically lower for CSPR, are offset by the broader dip and lower DA, making the overall performance of the proposed design much more efficient in light–matter interaction [7],[1]. The $S_{img.}$ further supports this, showing values of 18361 RIU⁻¹, 20320RIU⁻¹, and 22237 RIU⁻¹ for monolayer, bilayer, and tri-layer of ReS₂ respectively, all of which immensely exceed the $S_{img.}$ (2002 RIU⁻¹) of the CSPR. The CSPR sensor shows a PD of 225.14 nm, while the proposed sensor achieves significantly deeper PD of 713.11 nm for monolayer, 676.44 nm for bilayer, and 644.22 nm for trilayer of ReS₂. This deeper PD facilitates better interaction with analytes, particularly useful in biochemical and molecular sensing where analytes may not be strictly surface-bound.

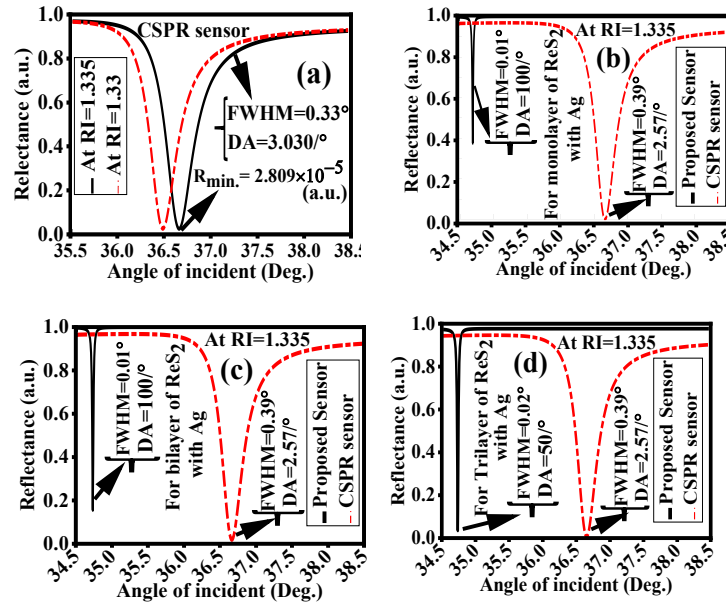


Fig.10 Performance parameters comparison of proposed and CSPR sensor with respect to (a) Reflectance vs Incident angle at RI=1.33 and 1.335 using CSPR sensor (b) Reflectance vs incident angle for monolayer of ReS₂ in the proposed sensor (c) Reflectance vs incident angle for bilayer of ReS₂ (d) Reflectance vs incident angle for tri-layer of ReS₂.

Table 7: Performance Comparison of the Proposed Sensor with CSPR Sensor at RI =1.335 of SM.

Types Of sensor	FWHM (°)	DA(1/°)	$S_{ang.}(^{\circ}/RIU)$	$FOM_{ang.}(RIU^{-1})$	$R_{min.}(a.u.)$	$\theta_t(^{\circ})$	$S_{img.}(RIU^{-1})$	IFOM $[(^{\circ}RIU)]^{-1}$	PD (nm)
CSPR sensor	0.39	2.57	36	92.31	2.809×10^{-5}	36.66	2002	5133.34	225.14
Proposed sensor with a monolayer of ReS ₂	0.01	100	12	1200	0.3761	34.69	18361	1836100	713.11
Proposed sensor with a bilayer of ReS ₂	0.01	100	12	1200	0.1393	34.70	20320	2032000	676.44
Proposed sensor with a trilayer of ReS ₂	0.02	50	12	750	0.03647	34.72	22237	1111850	644.22

Ultimately, the RC in Fig.10 and the performance parameters in Table 7 conclusively establish that the proposed one is superior to the CSPR sensor in terms of sharpness of resonance curve, DA, PD, sensitivity, and overall efficiency. Whether using a monolayer, bilayer and tri-layer of ReS₂, the sensor consistently achieves better results, making it a highly effective and reliable alternative to traditional SPR-based sensing platforms.

6. Fabrication Steps to develop chip of the proposed sensor

Under this section, the fabrication workflow to develop the chip of the proposed one have been shown in the Fig.11. In the step I ,The first layers of 2S2G prism with a high RI, is first cleaned with vapour of acetone and ionized water to make it for the deposition of DBL [1] [7]. In step II, the second layer, Cytop, is applied over the flat face of the prism by the spin coating method [1],[36], [37] with its thickness precisely controlled to 1500 nm by a quartz crystal monitor. Next, thin film of Ag is coated on the cytop coating using the physical vapour deposition (PVD) method [38] in the step III. The fourth layer, LiNbO₃, is then deposited on the Ag thin film using sol-gel fabrication method [39] in the step IV. Following this, the fifth layer, ReS₂ is applied over the LiNbO₃ using PVD [40] in step V.

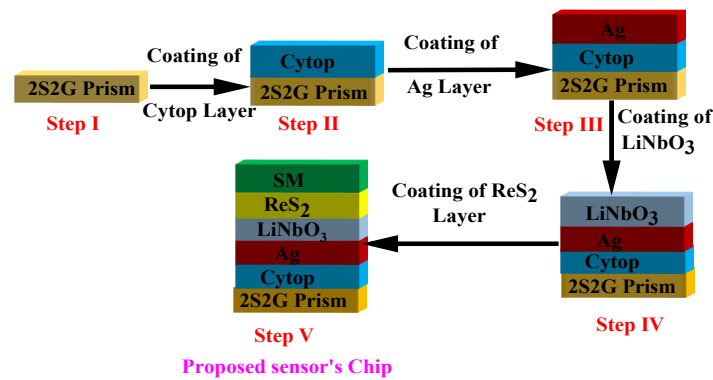


Fig.11 Fabrication workflow for developing the proposed sensor

7. Comparison of performance with the State-of-the-art

Table 8 clearly demonstrates that the proposed sensor surpasses the prism-coupled SPR and LRSPR sensors reported in the literature, which are mentioned in Ref. [28]to [29], [41] to[42] and Ref. [43] respectively,

regardless of whether ReS₂ is deployed as a monolayer, bilayer, and tri-layer. Whereas previous works achieve $S_{ang.}$ up to 388 °/RIU but suffer from very low $DA \leq 0.172^\circ$ and $FOM_{ang.} \leq 54 \text{ RIU}^{-1}$, but our sensor delivers a DA of 100° with monolayer and bilayer of ReS₂ and 50° with tri-layer, along with an $S_{ang.}$ of $12^\circ/\text{RIU}$. $S_{ang.}$ is multiplied by narrow FWHM (0.01° to 0.02°) to yield $FOM_{ang.}$ of $1,200 \text{ RIU}^{-1}$ (in case of monolayer/bilayer) and 750 RIU^{-1} (in case of tri-layer), which is higher than the best literature value of $FOM_{ang.}$ of 54 RIU^{-1} . $S_{img.}$ and IFOM further underscores the superiority of the sensor.

Table 8: Performance assessment of the proposed one against the existing sensor used for detection of Analyte of RI of 1.33-1.335

Reference No	Structure of Prism-coupled SPR/LRSPR sensor	$S_{ang.}$ (°/RIU)	DA (Deg. ⁻¹)	$FOM_{ang.}$ (RIU ⁻¹)	S_{img} (RIU ⁻¹)	IFOM [(Deg.RIU) ⁻¹]	PD (nm)	Analyte RI Range
[29]	BK7prism/Au/Ti ₃ C ₂ T _x (Mxene)/WS ₂ /BP/SM	190.22	0.0815	15.51	-	-	-	1.33-1.335
[41]	BK7 prism/Au/ Ti ₃ C ₂ T _x /SM	160	-	-	-	-	-	1.33-1.335
[44]	BK7prism/Ag/Si/BP/Mxene/SM	264	0.15	41.25	-	-	-	1.33-1.335
[45]	BK7 prism/Au/WSe ₂ /Au/Mxene/SM	198	-	-	-	-	-	1.33-1.335
[42]	BK7Prism/Ag/Si/Mxene/SM	231	0.172	39.83	-	-	-	1.33-1.335
[28]	BK7prism/Ag/BP/WS ₂ /Ti ₃ C ₂ T _x /SM	388	0.139	54.04	-	-	-	1.33-1.335
[43]	2S2G prism /Teflon/Ag/Mxene/SM	15.96	12.5	199.5	4998	62475	546.65	1.33-1.335
This work	2S2Gprism/Cytop/Ag/LiNbO ₃ /ReS ₂ (1*0.7nm)/SM	12	100	1200	18361	1836100	713.11	1.33-1.335
	2S2Gprism/Cytop/Ag/LiNbO ₃ /ReS ₂ (2*0.7nm)/SM	12	100	1200	20320	2032000	676.44	1.33-1.335
	2S2Gprism/Cytop/Ag/LiNbO ₃ /ReS ₂ (3*0.7nm)/SM	12	50	750	22237	1111850	644.22	1.33-1.335

The best existing LRSPR design in Ref. [43] reports $S_{img.}$ of $4,998 \text{ RIU}^{-1}$ and IFOM of $546.7 (\text{°RIU})^{-1}$, whereas our proposed sensor with monolayer, bilayer, and tri-layer of ReS₂ achieves $S_{img.}$ of $18,361 \text{ RIU}^{-1}$, $20,320 \text{ RIU}^{-1}$ and $22,237 \text{ RIU}^{-1}$, and IFOM of $1.8361 \times 10^6 (\text{°RIU})^{-1}$, $2.0320 \times 10^6 (\text{°RIU})^{-1}$ and $1.11185 \times 10^6 (\text{°RIU})^{-1}$, respectively. Finally, the PD in our designs (713.11 nm, 676.44 nm, and 644.22 nm) exceeds the 546.65 nm reported for the LRSPR sensor mentioned in Ref.[43]. Finally, these results illustrate that by incorporating a thin layer of LiNbO₃ as passivation layer and using the anisotropic properties of ReS₂, our proposed sensor surpasses the performance of both conventional and existing SPR and long-range SPR sensor across every key metric.

Conclusion

This proposed work shows that ReS₂ can be effectively used as a BRE in an LRSPR-based sensor. By utilising the anisotropic optical properties of ReS₂ and by combining it with a multilayer sensor structure, the proposed sensor achieves significant improvement in $S_{img.}$, angular resolution, and PD. The use of an Ag layer along with a protective LiNbO₃ coating provides strong electric field confinement and enhances interaction between the sensor surface and analyte molecules. Compared to CSPR sensors, the proposed LRSPR sensor can detect very small changes in RI over a larger sensing depth with better accuracy. In comparison with recently reported LRSPR sensors, the proposed design offers superior performance parameters and proves its practical usefulness. The

modelling and fabrication feasibility further support its application in real biosensing systems. Overall, this work presents a novel and high-performance LRSPR sensor design that can improve the sensitivity and accuracy of future optical biosensing technologies.

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Statements & Declaration

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Author Contributions

Virendra Kumar: Conceptualisation, design, modelling, Simulation, Formal analysis, Writing – original draft,

Sarika Pal: Supervision, Writing- revise draft

Vivek Singh: Supervision

Yogendra Kumar Parajapati: Supervision

Ethics approval is not required, as the data presented here is based on a numerical study.

Data Availability: The generated dataset is already presented in the manuscript.

Consent to Publish: All Authors express their consent to publish this manuscript.

Consent to participate: No consent is required.

Competing Interests: The authors have no competing interests.

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